OPEN SOURCE, NOVEMBER 2026

Teaching a small OCR model to read Italian documents

From the stock model to today, one step at a time, on a Mac with 16 GB of memory.

1.2 GB
the model, against 2.7 GB for the original

Local
runs on Apple Silicon with MLX, and with llama.cpp elsewhere

Decided first
every test is written down before it runs, losses reported next to gains

Four steps

1. **Original.** GLM-OCR by Z.ai, as published: a small vision-language model that turns a page image into text.

2. **Fine-tuned.** A small adapter trained on 91 Italian and English pages the original misread, then compressed to 8 bits: 45% of the size.

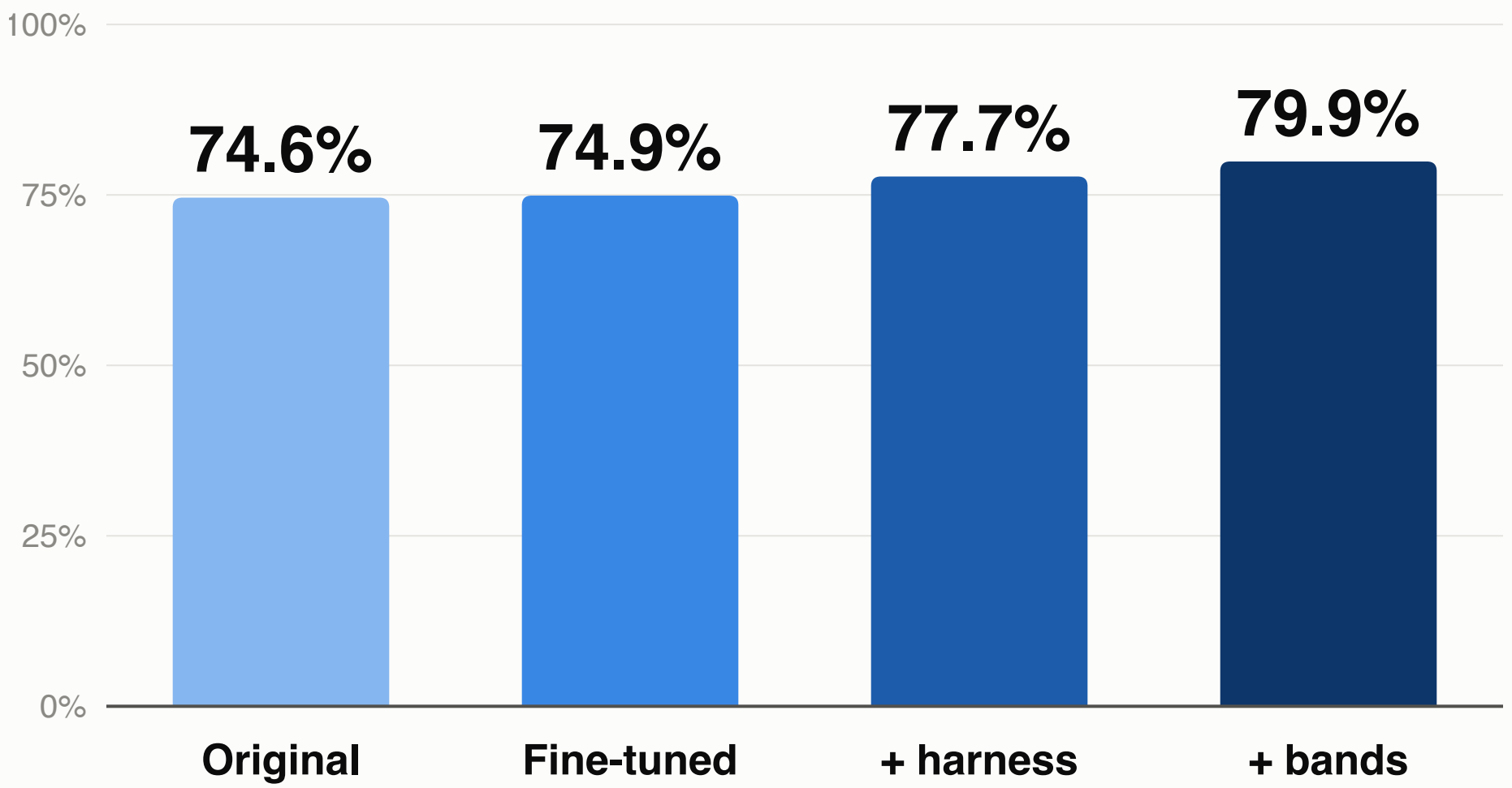
3. **+ harness.** The reader around the model. It stops the model when it starts repeating itself and reads the page again, and it reads a PDF's own text when that text can be trusted.

4. **+ bands.** The last two days, released in v1.5.0. The top and bottom of each page are read a second time, to catch letterheads and closing lines the whole page loses. A small orientation check skips pages lying on their side.

The score on every chart: the share of key fields read exactly right (amounts, dates, tax codes, IBANs, form fields), or the benchmark's own score.

Real Italian forms

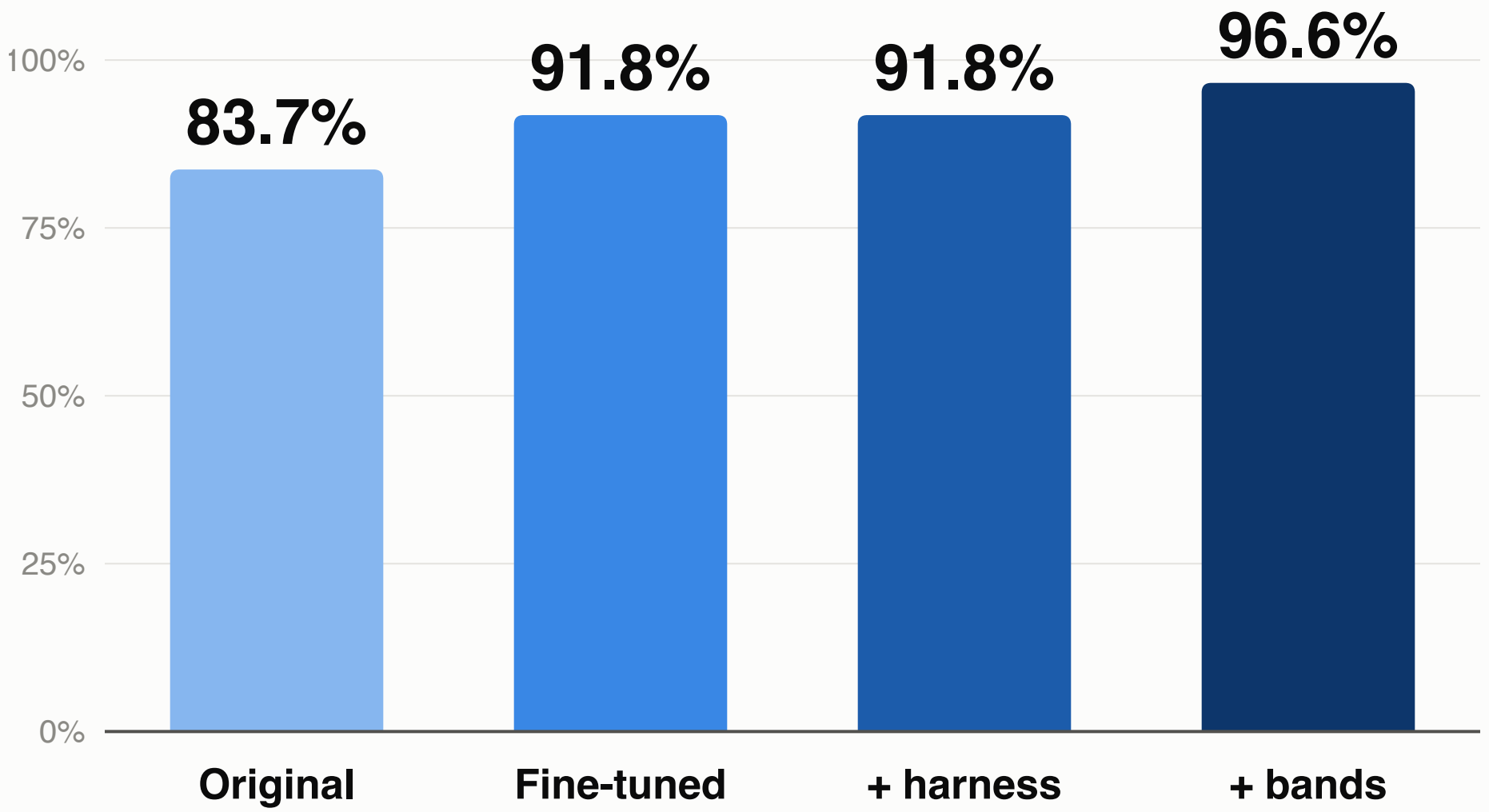
XFUND, 50 scanned forms, test set



Fields missed: **25.4% to 20.1%**. The fine-tuning alone was a tie here; the harness and the bands made the difference.

Stress pages

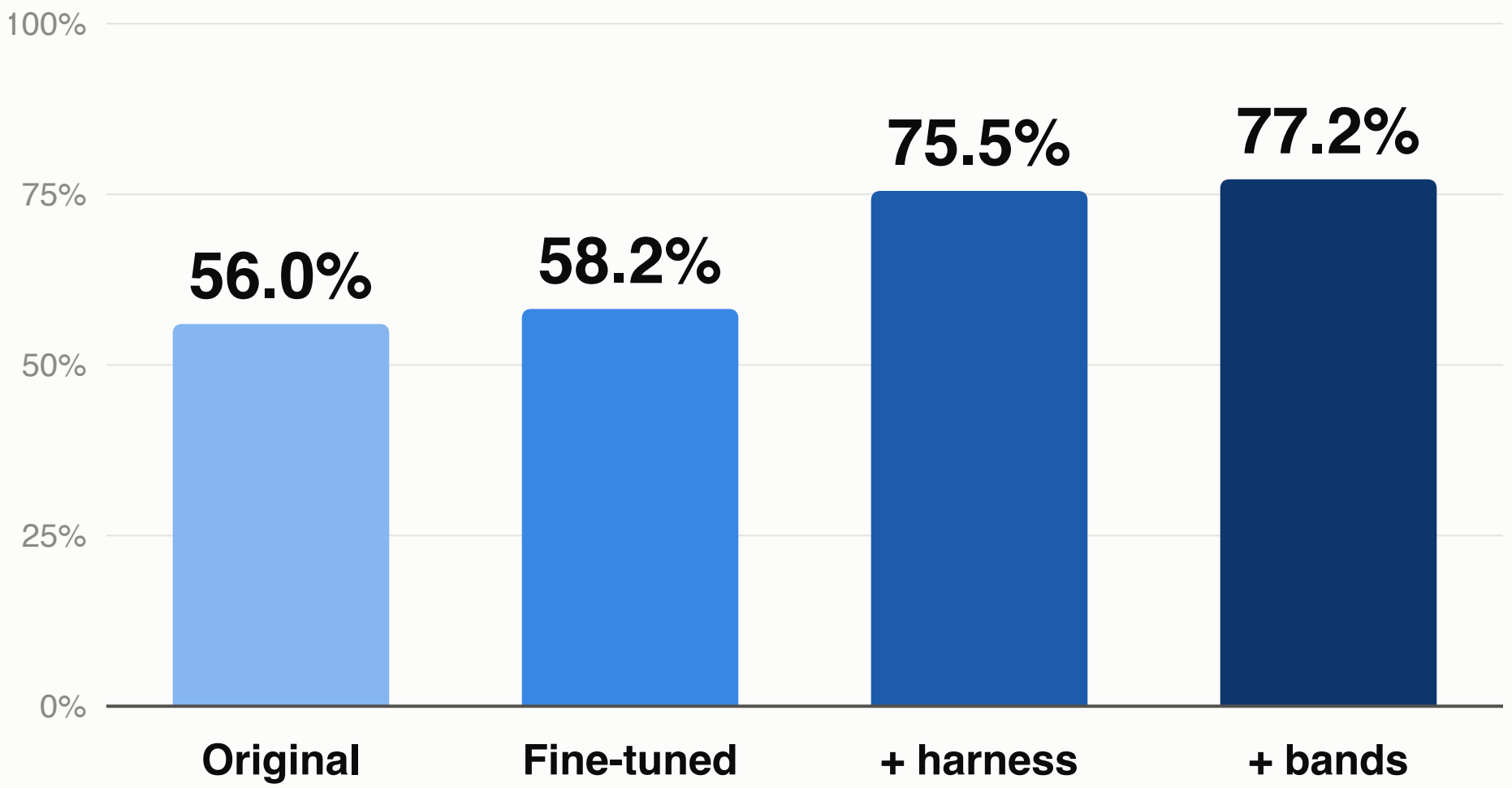
ocr-bench-v1, 50 pages: phone captures, degraded scans, stamps, Italian and English, test set



Fields missed: **16.3% to 3.4%**, almost five times fewer. The fine-tuning did most of it; no page loops here, so the harness had nothing to fix. No page got worse at any step.

Real PDFs

Omni Parse Bench by Datalab, 80 real PDF pages, the benchmark's own text tests



The big step is the harness reading the PDF's own text where it passes four checks, an idea taken from Datalab's Indra OCR. The bands then lift the pages the model reads, losing none.

What it costs, and what it does not do

- **Time.** The bands add 15 to 25% to the time of a page. On 200 letters no model had seen, they took entity recall from 92.4% to 96.1%, and the pages whose closing lines were lost from 16 to 1.
- **Memory.** 2.1 GB at peak, against 3.2 GB for the original.
- **Chinese** pages read whole are worse than with the original. Pages turned by 90 degrees still have to be straightened first.
- **How it is measured.** Every test is decided in a public ledger before it runs, with paired 95% intervals over pages, and every number is recomputed from the raw outputs by the test suite.

github.com/fabriziosalmi/glm-ocr-ita

huggingface.co/fabriziosalmi/GLM-OCR-ita-mlx-8bit

Enjoy and contribute.